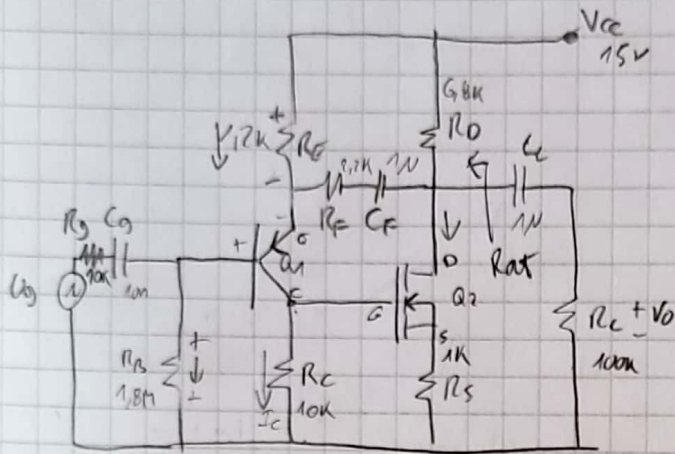




1) op point of Q_1 and Q_2

$$+R_1 I_B - V_{BE1} + R_5 I_E = V_{CC}$$

$$R_1 I_B + R_5 I_E (1 + \beta_{F1}) = V_{CC} + V_{BE1}$$

$$I_B = \frac{V_{CC} + V_{BE1}}{R_1 + R_5 (1 + \beta_{F1})} = 7,44 \mu A$$

$$I_C = 0,744 mA$$

$$V_{CE1} = V_C - V_E = R_C I_C - (V_{CE} - R_5 I_E) = -6,65 V$$

$$\begin{cases} I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \\ \frac{R_C I_C - V_{GS}}{R_S} = I_D \end{cases} \quad \frac{R_C I_C - V_{GS}}{I_{DSS} R_S} = 1 + \frac{V_{GS}^2}{V_T^2} - \frac{2 V_{GS}}{V_T}$$

$$V_{GS}^2 \left(\frac{1}{4}\right) + V_{GS} (4) + 36,2 = 0$$

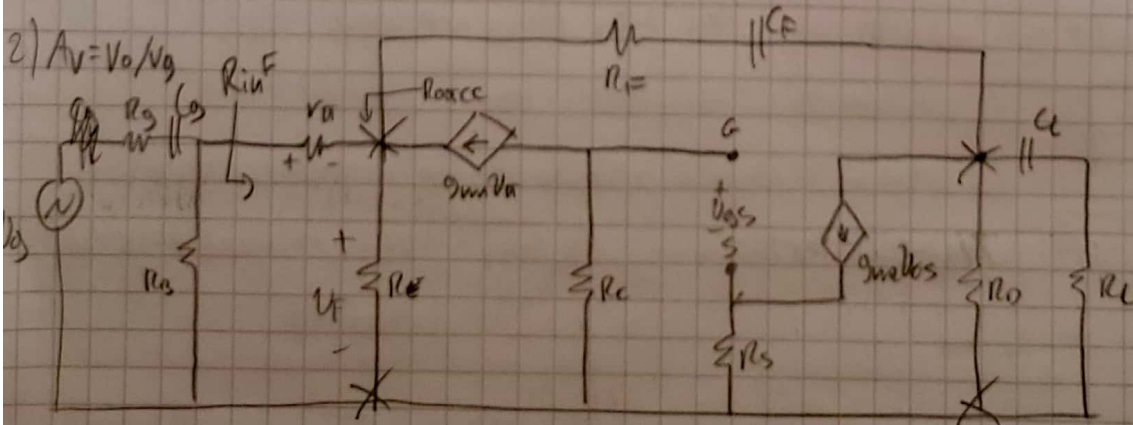
$$V_{GS12} = \begin{matrix} 6,65 V \\ -8,2 \end{matrix}$$

$$I_D = 0,99 mA \quad V_{DS} = V_{CC} - R_D I_D - R_S I_D = 7,28 V$$

$$Q_1 (-6,65 V, 0,744 mA) \quad Q_2 (7,28 V, 0,99 mA)$$

$$g_{m1} = 2,76 mA/V = 3,36 k\Omega$$

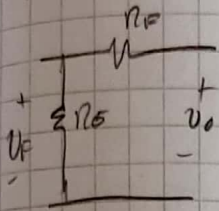
$$g_{m2} = 0,465 mA/V$$



short aip $\rightarrow$ no feedback $\rightarrow$ Voltage coupling

short input $\rightarrow$ feedback works $\rightarrow$ Voltage mixing

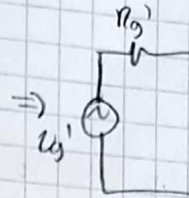
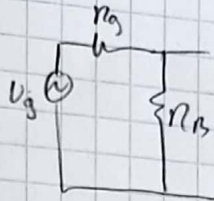
β network: $V_F = \beta V_O$



$$\beta = \frac{R_E}{R_F + R_E} = 0.356 \quad R_{if} = R_E // R_F = 0.776 \text{ k}\Omega \quad R_{of} = R_E + R_F = 3.4 \text{ k}\Omega$$

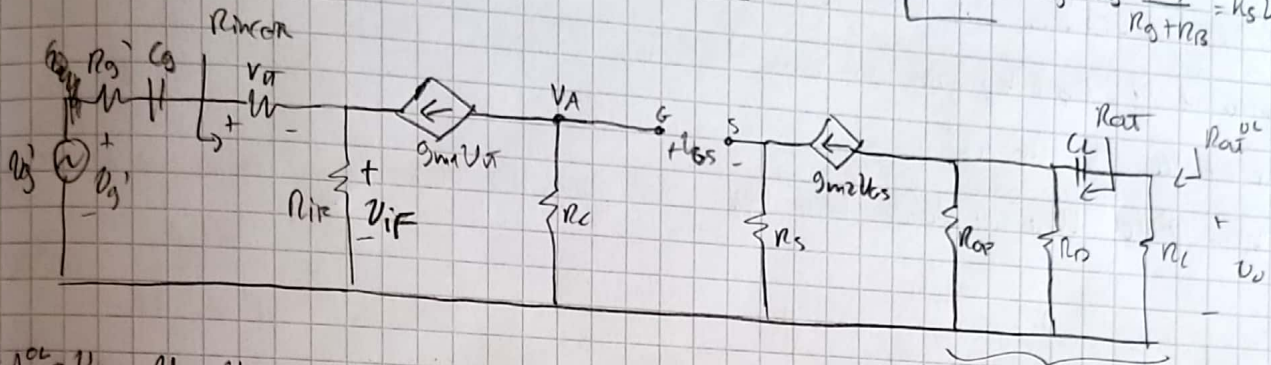
$$\begin{cases} V_F = \beta V_O + R_{if} i_{if} \\ R_O = V_O / i_{of} \end{cases}$$

theorem to create single loop



$$R_g' = R_g // R_h = 995 \text{ k}\Omega$$

$$V_g' = V_g \frac{R_L}{R_g + R_L} = 4.5 V_g$$



$$A_v^{OL} = \frac{V_O}{V_g} = \frac{V_O}{V_A} \cdot \frac{V_A}{V_g}$$

$$R_L' = R_{of} // R_O // R_L = 2.216 \text{ k}\Omega$$

$$R_{in}^{OL} = 81.736 \text{ k}\Omega$$

$$\frac{V_O}{V_A} = \frac{-g_{m2} V_{os} R_L'}{V_{os} + R_{os} g_{m2} V_{os}} = \frac{-g_{m2} R_L'}{1 + g_{m2} R_{os}} = -0.682 \text{ V/V}$$

$$\frac{V_A}{V_g'} = \frac{-g_{m1} V_A R_C}{(R_g' + r_{\pi}) \frac{V_A}{r_{\pi}} + R_{if} V_A \left(\frac{1 + \beta_0}{r_{\pi}} \right)} = \frac{-g_{m1} R_C r_{\pi}}{R_g' + r_{\pi} + R_{if} (1 + \beta_0)} = \frac{-\beta_0 R_C}{R_g' + r_{\pi} + R_{if} (1 + \beta_0)} = -10.9$$

$$A_{vol} = 7.438$$

$$A_F = \frac{A_v^{OL}}{1 + \beta A_v^{OL}} = 2.103$$

$$A_v = \frac{V_O}{V_g} = \frac{V_O}{V_g'} \cdot \frac{V_g'}{V_g} = A_v^F \cdot \frac{R_g}{R_g + R_B} = \boxed{2.027}$$

4) Root of fyne feedback decrease Root

$$R_{out}^a = R_L' = 2.216 \text{ k}\Omega \quad R_{out}^F = \frac{R_{out}^a}{1 + A_{vol} \beta} = \boxed{607 \text{ }\Omega}$$

5) low frequency response → directly from initial network, $R_{in}^F = R_{in}^a (A_{if})$

$$R_g^{sc} = R_g + R_B // (R_{in}^F - R_g) = \boxed{286 \text{ k}\Omega}$$

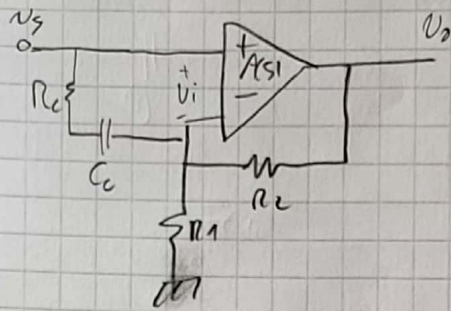
$$R_{if}^F = (R_g // R_L) + R_F + R_{out}^a / R_E = R_g // R_L + R_F + \frac{r_{\pi} + R_B // R_g}{1 + \beta_0} // R_E = 9.68 \text{ k}\Omega$$

$$R_{cl}^{sc} = R_{out}^F + R_L = 100.6 \text{ k}\Omega$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_F R_{if}^{sc}} + \frac{1}{C_g R_g^{sc}} + \frac{1}{R_{cl}^{sc} C_L} \right) = \boxed{76 \text{ Hz}}$$

E32

$$R_1 = 18 \text{ k}\Omega \quad R_2 = 56 \text{ k}\Omega$$



$$A(s) = \frac{A_0}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

$$\frac{10^2}{5 \cdot 10^5}$$

$$1) R_c \rightarrow \infty \quad \varphi_m$$

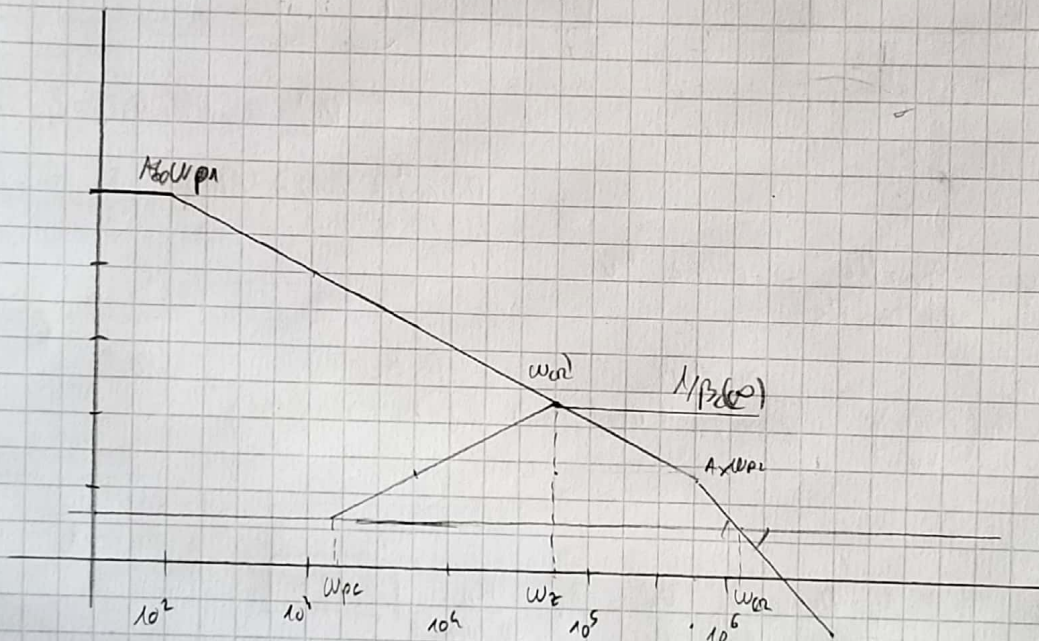
$$2) R_c \rightarrow \infty \quad \text{block diag (feedback trans)}$$

$$3) \text{ expression of closed loop gain considering } 1 \ll T$$

$$4) \text{ Value of } R_c \text{ for } \varphi_m = 45^\circ$$

Hint: place ω_{cr1} 1 decade before ω_{p2} and place $\omega_z = \omega_{cr2}$

1)



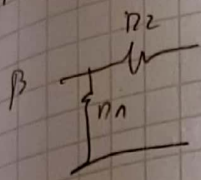
$$\Gamma \approx 1R = \frac{V_r}{V_t} = A(s) \frac{R_1}{R_1 + R_2} \quad \beta = \frac{R_1}{R_1 + R_2} = 0,2342 \quad \frac{1}{\beta} = 4,11 = 12,3 \text{ dB}$$

Notes: $G \propto \beta \rightarrow \varphi_m$ very low

$$A_0 \omega_{p1} = A_x \omega_{p2} \quad \left\{ \begin{array}{l} A_x = \frac{A_0 \omega_{p1}}{\omega_{p2}} \\ A_x \omega_{p2}^2 = \frac{1}{\beta} \omega_{cr}^2 \end{array} \right. \quad A_0 \omega_{p1} \omega_{p2} \beta = \omega_{cr}^2 \rightarrow \omega_{cr} = \sqrt{A_0 \omega_{p1} \omega_{p2} \beta} = 1,08 \cdot 10^6 \text{ rad/s}$$

$$\varphi_m = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_{p1}}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_{p2}}\right) = 24,8^\circ \quad \text{not that bad}$$

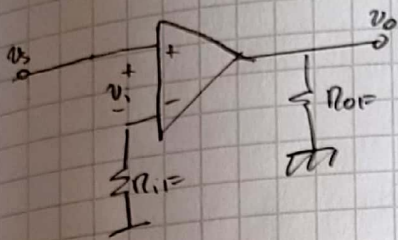
2) now inverting $\rightarrow$ voltage sampling voltage mixing



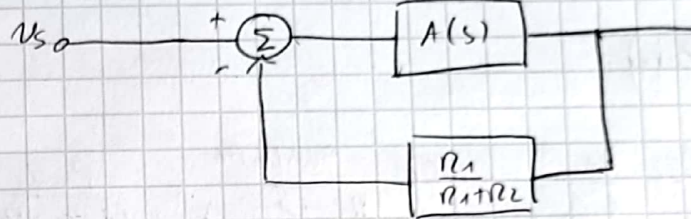
$$R_{iF} = R_1 \parallel R_2 = 13.62 \text{ k}\Omega$$

$$R_{oF} = R_1 + R_2 = 76 \text{ k}\Omega$$

$$\beta = \frac{R_1}{R_2 + R_1} = 0.2432$$



$$A_d = \frac{V_0}{V_1} = A(s)$$



3) closed loop exp with 1CCT

$$A^F(s) = \frac{A_d(s)}{1 + \beta A_d(s)} = \frac{A(s)}{1 + A(s) \cdot \frac{R_1}{R_1 + R_2}} \approx \frac{1}{\frac{R_1}{R_1 + R_2}} = \frac{R_1 + R_2}{R_1} = \frac{1}{\beta} = 12.3 \text{ dB}$$

4) value of R_C and C_C to get $\phi_m = 45^\circ$

$$\omega_{cr} = \frac{\omega_{p1}}{10} = 5 \cdot 10^4 \text{ rad/s}$$

$$\omega_2 = \omega_{cr}$$

$$\begin{cases} A_{d0} \omega_{p1} = \omega_2 / \beta_{100} \\ \frac{1/\beta_{100}}{\omega_{pc}} = \frac{1/\beta_{100}}{\omega_2} \end{cases} \Rightarrow \begin{cases} \frac{1}{\beta_{100}} = \frac{A_{d0} \omega_{p1}}{\omega_2} = 200 \\ \omega_{pc} = \frac{1/\beta_{100}}{1/\beta_{100}} \cdot \omega_2 = 1.027 \cdot 10^3 \text{ rad/s} \end{cases}$$

$$\beta_C(\omega) = \frac{R_1 R_C}{R_1 \parallel R_C + R_2} = \frac{\frac{R_1 R_C}{R_1 + R_C}}{\frac{R_1 R_C}{R_1 + R_C} + R_2} = \frac{\frac{R_1 R_C}{R_1 + R_C}}{\frac{R_1 R_C + R_2 R_1 + R_2 R_C}{R_1 + R_C}} = \frac{R_1 R_C}{R_C(R_1 + R_2) + R_2 R_1} = \frac{1}{200}$$

$$R_1 R_C = R_C \left(\frac{R_1 + R_2}{200} \right) + \frac{R_1 R_2}{200}$$

$$R_1 R_C - R_C \left(\frac{R_1 + R_2}{200} \right) = \frac{R_1 R_2}{200}$$

$$R_C \left(R_1 - \frac{R_1 + R_2}{200} \right) = \frac{R_1 R_2}{200}$$

$$R_C = 285 \Omega$$

$$C_C = \frac{1}{\omega_{cr} R_C} = 70.17 \text{ nF}$$

2h